

Forecasting Realized Volatility: Does Anything Beat Linear Models?

Rafael R. Branco ^{*} Alexandre Rubesam [†] Mauricio Zevallos [‡]

September 23, 2022

Abstract

We evaluate the performance of several linear and nonlinear machine learning models in forecasting the realized volatility (RV) of ten global stock market indices in the period from January 2000 to December 2021. We train models using a dataset which includes past values of the RV and additional predictors (composed by lagged returns and macroeconomic variables), and compare them to the widely used heterogeneous autoregressive (HAR) model. Our main conclusions are that (i) the additional predictors improve the out-of-sample forecasts of 1-day-ahead RV, and (ii) we find no evidence that nonlinear models can outperform statistically either linear models or HAR models with additional predictors.

Keywords: Volatility forecasting, Machine Learning, Realized Volatility, Model Confidence Set, Value-at-Risk.

JEL classifications: C22, C52, C53, C58, G15.

^{*}Graduate Program of Statistics, University of Campinas, Brazil. Email: ra8branco@gmail.com

[†]Corresponding author. Department of Finance, IESEG School of Management, France. E-mail: a.rubesam@ieseg.fr

[‡]Department of Statistics, University of Campinas, Brazil. Email: amadeus@unicamp.br

1 Introduction

Forecasting asset return volatility is a central problem in many financial applications, including the pricing and hedging of derivatives, risk management, and asset allocation. A large variety of parametric volatility models have been proposed to capture stylized features of financial asset returns, such as asymmetric responses in volatility to innovations in returns (leverage effect), the effect of volatility on returns (volatility feedback), and long memory, among others.¹ One challenge with evaluating these models is that volatility is unobservable. Over the last two decades, however, several *realized volatility* (henceforth, RV) estimators, which make use of high-frequency (i.e., intra-day) data, have been developed ([Andersen et al., 2001b](#); [Meddahi, 2002](#); [Barndorff-Nielsen and Shephard, 2002](#)). These consistent estimators of the true volatility are often considered as suitable benchmarks to assess the accuracy of forecasts from parametric volatility models. In addition, direct time series modeling of RV outperforms GARCH and stochastic volatility models in terms of out-of-sample accuracy ([Andersen et al., 2001b, 2003](#)).

[Corsi \(2009\)](#) proposed a parsimonious model to forecast RV, the heterogeneous autoregressive (HAR) model, in which RV is regressed onto lagged moving averages representing different horizons (i.e. daily, weekly, and monthly). Despite its simplicity, the HAR model provides very competitive RV forecasts. Given the complex, nonlinear dynamics of RV, an interesting question is whether adopting a more flexible modeling approach, such as those offered by machine learning (ML) models, can lead to improved forecast accuracy. In addition to providing flexibility and letting the models learn the complex dynamics from past data, this approach can also exploit information from additional predictors, such as macroeconomic variables. Although

¹Some prominent GARCH-type models that deal with these stylized facts include the ([Glosten et al., 1993](#)) (GJR) GARCH and the exponential GARCH (EGARCH) model of [Nelson \(1991\)](#) (leverage effect); the GARCH-in-mean model of [Engle et al. \(1987\)](#) (volatility feedback); the Integrated (IGARCH) model of [Engle and Bollerslev \(1986\)](#) and the Fractionally Integrated (FIGARCH) model of [Baillie et al. \(1996\)](#) (long memory). See [Andersen et al. \(2010\)](#) and [Teräsvirta \(2009\)](#) for surveys of GARCH and stochastic volatility models.

ML models have been extensively used in many financial applications, their use in volatility forecasting is still relatively limited and, to our knowledge, a comprehensive evaluation of the role of macroeconomic variables in forecasting high-frequency RV is still missing in the literature.

In this paper, we perform an empirical comparison of the performance of different statistical methods in forecasting the RV volatility of ten global stock market indices. We compare the performance of the HAR model with that of different ML models trained on an augmented set of predictors, which includes those used in the HAR model, as well as lagged returns and various macroeconomic variables. We consider two different RV measures, and train models to forecast 1-day, 5-day, and 22-day-ahead RV. We train linear models with and without regularization (Lasso, Ridge regression, Elastic net), as well as nonlinear ML models such as Random Forests, Boosting, and different types of neural networks. The performance of the methods is assessed through two criteria: the out-of-sample root mean squared error (RMSE) and the Value-at-Risk (VaR). For each stock index, we compare the performance of different ML models through the model confidence set (MCS) procedure of [Hansen et al. \(2011\)](#). For a given confidence level, the MCS procedure generates a set of superior models (SSM), for which the hypothesis of equal forecasting performance cannot be rejected. This methodology sheds light on our two main research questions, (i) are the additional predictors useful in forecasting RV? and (ii) are nonlinear ML models superior to the simpler, linear models in forecasting RV? To the best of our knowledge, our article is the first one to systematically study the performance of a different ML models in forecasting the high-frequency RV of a large group of stock indices, using a wide set of predictors.

The main contributions of this paper can be summarized as follows. First, we show that including additional predictors improves the (out-of-sample) forecasting performance of 1-day-ahead RV for all indices, relative to the HAR model. For all indices, an augmented HAR model with functions of lagged returns and macroeconomic variables (HARX) produces lower out-of-sample RMSE relative to the HAR. In

addition, the HAR model is often excluded from the SSM, whereas the HARX model is always included. However, we find that the ability of the additional predictors to improve the forecasting performance is limited to the 1-day forecast horizon. For 5-day and 22-day forecast horizons, the HAR is usually the best performing model in terms of RMSE and is always included in the SSM.

Second, we find no evidence that nonlinear ML models can outperform linear ones to forecast high-frequency RV. Even when a nonlinear ML model produces the lowest out-of-sample RMSE, other linear models achieve similar results and are included in the set of superior models. This result is in contrast with some previous studies, which find that ML models significantly outperform linear models to forecast stock index RV estimated from daily returns, or individual stock RV using high-frequency data.

Finally, we investigate the usefulness of RV forecasts from different ML models to estimate Value-at-Risk (VaR). The results suggest that VaR forecasts based on forecasts of the realized kernel variance (two-scale/Bartlett) RV measure of [Barndorff-Nielsen et al. \(2008\)](#) are better than those obtained using the 5-minute RV measure, but overall, several methods yield similar performance. However, we note a general tendency to underestimate the VaR, i.e. the frequency of observed VaR violations typically exceeds the expected VaR level.

The remainder of this paper is structured as follows. Section 2 presents a short review of the literature on volatility forecasting using machine learning. Section 3 presents the data set. Section 4 describes the methodology; the considered Machine Learning methods, the fitting procedure, the metrics to compare the models and computing details. The results and their discussion are presented in Section 5. Section 6 concludes.

2 Literature Review

There is a vast literature investigating the use of ML models in a variety of financial applications. A large part of this literature focuses on forecasting returns or their

directions. Some examples include [Sermpinis et al. \(2013\)](#) (forecasting changes in exchange rates), [Huck \(2009, 2010\)](#) (forecasting returns to develop statistical arbitrage strategies), [Kaucic \(2010\)](#) (development of trading systems using technical indicators), and [Akyildirim et al. \(2020\)](#) (prediction of the direction of cryptocurrency prices).²

Recently, there has been a surge of interest in ML within the finance and economics fields, which can be explained by the success of ML in different domains, as well as the fact that typical econometric techniques and models are not necessarily well-suited to deal with the vast amount of data that have become easily available to researchers and practitioners.³ Many recent papers in this literature focus on using ML concepts, such as regularization and dimensionality reduction, to digest the large number of return predictors and identify drivers of asset return predictability.⁴ For example, [Gu et al. \(2020\)](#) and [Bianchi et al. \(2020\)](#) deploy a variety of ML models to forecast individual stock and bond returns, respectively, finding strong evidence of predictability. Both studies suggest that neural networks (NNs) perform best among a variety of ML models to forecast returns or yields.

Several studies have focused on using ML models to forecast the volatility of different types of assets. Many earlier studies, such as [Donaldson and Kamstra \(1997\)](#) and [Hu and Tsoukalas \(1999\)](#), focused on hybrid or semiparametric models that embedded models such as NNs into GARCH-style volatility models. [Schittenkopf et al. \(2000\)](#) follow a similar path, but employ a recurrent mixture density network. Interestingly, this paper uses high-frequency RV estimates as the measure of the “true” volatility to compare the accuracy of various models. While these studies sought to enhance GARCH models by directly introducing flexible NNs to model the dynamics of volatility, other studies have relied on using volatility estimates from

²See also [Atsalakis and Valavanis \(2009\)](#); [Henrique et al. \(2019\)](#), and [Sezer et al. \(2020\)](#) for surveys on the use of ML for prediction of financial time series.

³See [Varian \(2014\)](#), [Mullainathan and Spiess \(2017\)](#), and [Hsu et al. \(2016\)](#) for discussions about the role of ML in the econometrics toolbox, and comparisons between studies that use ML with those using linear econometric techniques.

⁴See for example [Feng et al. \(2020\)](#); [Freyberger et al. \(2020\)](#); [Kelly et al. \(2019\)](#); [Kozak et al. \(2020\)](#); [Lettau and Pelger \(2020\)](#).

standard GARCH-type models as predictors to train NNs, using RV measures as the target. For example, [Hajizadeh et al. \(2012\)](#) use volatility estimates from a pre-selected GARCH model as inputs to train a NN with additional variables to predict the monthly RV of the S&P 500 index, calculated with daily returns. A similar paper is [Kristjanpoller et al. \(2014\)](#), who use GARCH estimates as inputs in NNs to forecast the RV of three latin-american stock indices (calculated using daily returns), and found that the hybrid approach outperforms the GARCH model.

[Hamid and Iqbal \(2004\)](#) use NNs to forecast the RV of the S&P 500 index, measured on a monthly basis using daily returns. They compare the volatility forecasts from NNs with implied volatility, concluding that volatility forecasts from the NNs are superior. [Hillebrand and Medeiros \(2010\)](#) consider the application of Bagging ([Breiman, 1996](#)) to extensions of the HAR model that allow different lag orders and exogenous variables, with or without the inclusion of a NN to allow increased flexibility. They use the model to predict the RV of individual U.S. stocks, measured using the [Barndorff-Nielsen et al. \(2008\)](#) estimator, and find that both types of models benefit from Bagging. [Fernandes et al. \(2014\)](#) consider a similar approach to model the dynamics of the VIX index, and find that the linear HAR model performs as well as the more flexible approach with NNs, concluding that persistence is the only relevant feature for short-term forecasts of the VIX.

[Rosa et al. \(2014\)](#) use an evolving hybrid neural fuzzy network to forecast the 1-minute RV of stock indices, concluding that it outperformed a HAR benchmark, and performed similarly to a NN model. [Maciel et al. \(2016\)](#) develop a hybrid evolving fuzzy-GARCH approach and apply it to model the volatility of the S&P 500 and the Brazilian IBOVESPA indices. [Arrieta-Ibarra and Lobato \(2015\)](#) test the performance of regression trees, Random Forests, Support Vector Machines (SVMs), and NNs to predict both the returns as well as the volatilities (estimated using daily returns) of the S&P 500 index. They find no evidence of predictability of returns, and conclude that SVMs and NNs outperform a GARCH model for volatility forecasting. [Vortelinos \(2017\)](#) compares the HAR model with NNs, GARCH and the Principal Components

Combined (PCC) model to forecast RV (estimated using high-frequency data) in seven selected U.S. financial instruments from different asset classes.⁵ They conclude that the nonlinear models were not able to outperform the HAR. [Miura et al. \(2019\)](#) predict the RV of Bitcoin (calculated with 1-minute sampling) using NNs, Long Short-Term Memory (LSTM), SVM, Ridge regression and HAR, concluding that the linear Ridge regression model was the best performing model.

[Bucci \(2020b\)](#) consider four types of Recurrent Neural Networks (RNNs) to forecast the monthly RV (estimated using daily returns) of the S&P 500 index, concluding that they outperform time-series models such as HAR and ARFIMA. [Bucci \(2020a\)](#) consider a similar setup to estimate the realized covariance matrix of the S&P 500 index, the 10-year Treasury note future and the 1-month Treasury bond future, again finding superior performance for the RNN models. [Filipović and Khalilzadeh \(2021\)](#) predict the monthly RV of individual stocks (returns sampled at the daily frequency) using different ML models and a variety of individual firm characteristics as predictors. They find that an LSTM model with only lagged returns and volatilities as predictors provide the best performance, which is on par with models that include all the predictors. [Li and Tang \(2021\)](#) forecast the RV of individual stocks (estimated with high-frequency data) using a large number of predictors which include functions of past RV suggested by competing parametric models of RV, including the HAR model, and implied volatility measures from options data. In terms of ML models, they consider two linear models (Lasso and principal components regression) and three nonlinear models (Random Forests, gradient boosted regression trees, and NNs). The models are estimated by pooling the data of all the available stocks. Their results show that all ML models deliver relevant improvements relative to a HAR baseline (i.e. without any exogenous variables), and a simple ensemble that averages forecasts from the individual ML models works well to forecast RV across different forecast horizons. In a related study, [Zhang et al. \(2022\)](#) use pooled stock data to investigate

⁵The Dow Jones Industrial Average index, the EUR/USD exchange rate, the PowerShares QQQ exchange-traded fund, the E-Mini Dow futures continuous contract, the 30-year US Treasury yield, the Crude oil mini futures continuous contract, and the CBOE gold index options.

commonalities in intraday RV, and show that ML models outperform linear models in terms of forecasting power.

This short review of the literature shows a remarkable diversity of approaches and results. One commonality is that almost all papers surveyed rely on simple estimates of monthly RV using daily returns, which are much less precise when compared with the state-of-the-art high-frequency RV estimators mentioned previously. Among the few papers that use high-frequency RV estimates, two papers consider the RVs of individual stocks and find that ML is beneficial to improve RV forecasts. However, the papers employ completely different ML approaches (Bagging to select lags in modified NN-HAR models for each stock in [Hillebrand and Medeiros \(2010\)](#); different ML models estimated with pooled data for all stocks in [Li and Tang \(2021\)](#)), rendering any comparison or conclusion difficult. The remaining papers ([Vortelinos, 2017](#); [Rosa et al., 2014](#)) are also not comparable, as they differ in the choice of asset classes, the types of ML models, and the types of RV estimators.

3 Data and Realized Volatility

3.1 RV Measures

In this section, we briefly review the RV measures based on intraday sampling of returns. Assume the following process for the evolution of log prices of a given asset along day t :

$$dp(t + \tau) = \mu(t + \tau)d\tau + \sigma(t + \tau)dW(t + \tau), \quad 0 \leq \tau \leq 1, \quad t = 1, 2, \dots, \quad (1)$$

where $p(t + \tau)$ is the logarithmic price at time $t + \tau$, $\mu(t + \tau)$ is the drift component, $W(t + \tau)$ is a standard Brownian motion, and $\sigma(t + \tau)$ is the instantaneous volatility, which is assumed to be independent from $W(t + \tau)$. The *integrated variance* of the asset is defined by

$$IV_t = \int_0^1 \sigma^2(t + \tau - 1)d\tau, \quad (2)$$

and is considered a measure of the true day- t ex-post volatility.

Although the IV is a theoretical measure and is not directly observable, it can be consistently estimated using the sum of intraday squared returns ([Andersen, Bollerslev, Diebold, and Labys, 2003](#); [Barndorff-Nielsen and Shephard, 2002](#)). The estimators derived from this important result are generally referred to as *realized volatility* estimators.⁶ There are several approaches to sample intraday returns to estimate the IV (see [McAleer and Medeiros, 2008](#), for a review). We focus on two widely used estimators. The first one is the 5-minute RV estimator (*rv5*), which is the seminal measure discussed in [Andersen et al. \(2001a\)](#) and [Andersen et al. \(2003\)](#), and is defined as the sum of squared intradaily returns, sampled at the 5-minute frequency. In the absence of microstructure noise, this is an efficient estimator. Despite its simplicity, [Liu et al. \(2015\)](#) has shown that it is difficult to significantly beat the 5-minute RV estimator. The second estimator is the realized kernel variance (two-scale/Bartlett) (*rvts*), proposed by [Barndorff-Nielsen et al. \(2008\)](#), a more sophisticated approach which yields a consistent estimator in the presence of microstructure effects. This estimator was shown to be the top performer among 19 RV estimators in an extensive simulation study by [Gatheral and Oomen \(2010\)](#).

3.2 Description of the Data

We obtain daily RV estimates (*rv5* and *rvts*) for the ten stock market indices shown in Table 1 from the Realized Library of the Oxford-Man Institute of Quantitative Finance.⁷ The data cover the period from January 2000 to December 2021, and therefore include several periods of increased volatility or market stress, such as the Global Financial Crisis (2007-2008), the European sovereign debt crisis (from 2009 to late 2010s, peaking in 2012), the Chinese stock market bubble (2015-2016), Brexit (2016-2020), and the coronavirus pandemic from 2020.

[Table 1 around here]

⁶There is some variation in terminology in the literature, with some authors using the term “realized volatility” to refer to either an estimate of the integrated variance, its logarithm, or its square root. In this article, we use the latter.

⁷<https://realized.oxford-man.ox.ac.uk/>

We consider several predictors to train models to forecast RV, which are listed in Table 2. First, we include the three moving averages of past RV used in the HAR model of Corsi (2009), which correspond to horizons of 1, 5, and 22 days. These are denoted by $RV_t^{(n)}, n = 1, 5, 22$ (see Section 4.1.1). Next, we consider additional predictors which may contain information about the state of the markets, such as moving averages of the returns on the index ($r_t^{(n)}$), calculated at the same time horizons used in the HAR model (i.e., $n = 1, 5, 22$), as well as their squares ($[r_t^{(n)}]^2$). We also include, for each country: the daily change in the short-term and long-term benchmark interest rates (ΔSTR_t and ΔLTR_t , respectively);⁸ the daily change in the index dividend yield and earnings-to-price ratio (ΔDP_t and ΔEP_t , respectively); the exchange rate of the country currency relative to the US dollar (FX_t);⁹ the daily return of the S&P GSCI Commodity Index ($Comm_t$); and the daily change in the VIX volatility index (ΔVIX_t). Considering all the predictors described above, each model (with the exception of the HAR) includes 16 predictors.

It is important to highlight that some of the predictors above contain information pertaining only to each market, whereas other variables are global in scope or contain information related to specific markets (e.g., the change in the VIX) or the relationship between markets (e.g., the daily return of the country exchange rate, FX_t). The daily return of the S&P GSCI Commodity Index is included to investigate the potential impact of changes in the price of commodities, particularly on emerging markets such as Brazil and China. We justify the inclusion of the change in the VIX because this is a widely known, forward-looking implied volatility index of the largest financial market in the world, the United States.

[Table 2 around here]

⁸For most countries, the short-term and long-term benchmark rates are, respectively, the 3-month and 10-year government benchmark rates available on Refinitiv. The exceptions (due to data availability) are China, for which we used the 1-year government benchmark rate as the short-term rate, and Brazil, for which we use an overnight interbank rate as the short-term benchmark rate, following the convention in that market.

⁹For the United States, we use the dollar index (DXY).

4 Methodology

In this section we describe the methodology adopted in the paper. First, we present the ML models considered in the paper, as well as the fitting procedures and some computational details. Second, we present the metrics adopted for model comparison in terms of volatility forecasting ability.

4.1 Models

The general form of the predictive models considered in this paper is

$$RV_{i,t+k} = f(\mathbf{x}_{i,t}, \mathbf{z}_t; \boldsymbol{\beta}) + \epsilon_{i,t+k}, \quad (3)$$

where $RV_{i,t+k}$ is the k -period-ahead realized volatility of stock index i ; $\mathbf{x}_{i,t}$ is a vector containing the usual (index-specific) HAR predictor variables, e.g., moving averages of past RV corresponding to the daily, weekly, and monthly frequencies; $\mathbf{z}_t$ contains the additional predictors discussed in the previous section, and $\epsilon_{i,t+k}$ is an error term. The function f is the model-specific functional relating the predictors to the response, which depends on a vector of unknown parameters $\boldsymbol{\beta}$ to be estimated. For clarity, we omit the i subscript in the description of each ML model.

4.1.1 Heterogeneous Autoregressive Model of Realized Volatility (HAR)

Our benchmark model is the Heterogeneous Autoregressive Model for Realized Volatility (HAR) model proposed by [Corsi \(2009\)](#), which uses three moving averages of lagged RV, corresponding to the daily (one day), weekly (five days), and monthly (22 days) horizons. Despite its simplicity, the HAR model achieves a remarkable performance in terms of forecasting RV, and is a difficult benchmark to beat ([Patton and Sheppard, 2015](#)). The HAR model forecast is

$$RV_{t+k} = \beta_0 + \beta_d RV_t^{(1)} + \beta_w RV_t^{(5)} + \beta_m RV_t^{(22)} + \varepsilon_{t+k}, \quad (4)$$

where $RV_t^{(n)}$ is the n -day moving average of RV_t , i.e.

$$RV_t^{(n)} = \frac{1}{n} (RV_t + RV_{t-1} + \dots + RV_{t-n+1}). \quad (5)$$

In the context of equation (3), the HAR model corresponds to $f(\mathbf{x}_{i,t}, \mathbf{z}_t; \boldsymbol{\beta}) = \mathbf{x}_t \boldsymbol{\beta}$, with $\mathbf{x}_t = [1 \quad RV_t^{(1)} \quad RV_t^{(5)} \quad RV_t^{(22)}]$. Note that the HAR model omits the vector of additional predictors $\mathbf{z}_t$. The HAR model is the simplest one we consider and is estimated via OLS. All other models include additional predictors.

4.1.2 HAR Model with additional predictors (HARX)

When additional predictors are included in the HAR model (4), we refer to the model as the HARX model, which corresponds to the following equation:

$$RV_{t+k} = [\mathbf{x}_t \quad \mathbf{z}_t] \boldsymbol{\beta} + \varepsilon_{t+k}. \quad (6)$$

The HARX model is also estimated via OLS.

4.1.3 HARX with Ridge, Lasso, or Elastic Net penalty

Given that the HARX specification (6) is a linear model, including all variables may lead to poor out-of-sample performance (i.e., overfitting). Because of this, we consider three penalized linear methods: Lasso, Ridge and Elastic Net (Enet). A description of these methods can be found in Efron and Hastie (2016). These linear models have the same functional form as HARX, but are estimated by minimizing a penalized loss function:

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \left\{ RSS + \alpha_1 \sum_{j=1}^p (\alpha_2 \beta_j^2 + (1 - \alpha_2) |\beta_j|) \right\}, \quad (7)$$

where $RSS = \sum_{t=1}^T (RV_{t+k} - f(\mathbf{x}_{i,t}, \mathbf{z}_t; \boldsymbol{\beta}))^2$ is the residual sum of squares, α_1 is the elastic net penalty, and α_2 is a parameter controlling the degree of shrinkage. The Ridge regression model corresponds to $\alpha_2 = 1$ and the Lasso is obtained with $\alpha_2 = 0$. We also estimate the Elastic Net model, which is a compromise between the two, by using $\alpha_2 = 0.5$. In all cases, the optimal value of α_1 is chosen via cross-validation.

4.1.4 Tree-based Methods

The linear models described so far impose a strong restriction on the response function f , i.e., a linear relationship between the response variable and the predictors. Tree-

based methods provide a simple and powerful way to capture potential nonlinear relationships and interactions between predictors. These methods work by aggregating many regression trees, each of which is a simple, nonparametric model that works by partitioning the feature space into a set of distinct and nonoverlapping rectangular regions. The predicted value of a regression tree within each of these rectangles is the average of the observations in that region.

We consider three types of tree-based methods. The first two methods, bootstrap aggregation (or Bagging) and Random Forest, work by repeatedly bootstrapping a number B of samples from the data and training independent regression trees on each bootstrap sample. The forecast from these models is obtained as the average of the B forecasts from each individual tree. The main difference between Bagging and Random Forests is that, in the latter, a random subset of the total number of predictors is selected to estimate each tree, which tends to decrease the correlation between forecasts from individual trees. The hyperparameters in these models control the size of each individual tree, the number of variables to consider when training each tree (for the Random Forest model), and the total number of trees to estimate. Both methods are described in standard textbooks such as [Efron and Hastie \(2016\)](#).

The remaining tree-based method we use is Gradient Boosting, which, unlike Bagging and Random Forests, works by sequentially training shallow trees (called *weak learners*) on modified versions of the data. At each iteration of the algorithm, a new tree is trained on residuals from the previous iteration, i.e., each tree attempts to improve on its predecessors. The hyperparameters of this model include the number of splits of the tree used in each iteration, the total number of trees, and a learning rate, which controls the contribution of each tree added to the ensemble (see [Efron and Hastie, 2016](#)).

4.1.5 Neural Networks

The final category of nonlinear model we consider are neural networks with topologies similar to those in [Gu et al. \(2020\)](#) and [Rubesam \(2022\)](#), i.e., networks with one (NN1)

to five (NN5) hidden layers, with decreasing number of neurons in each subsequent hidden layer. The number of neurons in the first hidden layer is 32, and the number of neurons in each subsequent layer (if any) is half the number of neurons in the previous layer. Thus, NN1 has 32 neurons in its single hidden layer, NN2 has 32 neurons in the first layer and 16 in the second layer, and so on.¹⁰

4.2 Fitting Procedures

For each specific stock market index, let $r_1, \dots, r_T$ be the time series of returns and let $\sigma_1, \dots, \sigma_T$ denote the time series of realized volatilities (using either of the RV measures described earlier). Additionally, let $\hat{\sigma}_{t+k/t}$ denote the forecast of volatility at time $t+k$ based on the information up to time t ; thus k is the forecasting horizon. In this paper we considered $k = 1, 5, 22$, representing forecasts for one day ahead, one week ahead and one month ahead, respectively. With the exception of the HAR model, all models described in section 4.1 were fitted including the predictors listed in Table 2.

We train all competing models using an expanding training window approach. We start with a training window of six years, which is used to train the first batch of models, and an out-of-sample or testing window of one year. Once all models are fitted in the training window, we forecast on the testing window and store estimates and out-of-sample errors for further evaluation. Then, the training window is expanded to encompass the previous training and testing period, and a new one-year testing period is defined which begins the day after the last day of the new training period. This process is repeated until all available data is part of one of two periods. Thus, years 2000 to 2005 constitute the initial training period and we used testing periods of one year: 2006, 2007, $\dots$, 2021. Figure 2 depicts the process of dynamic windows used in the estimation.

¹⁰We have also tested LSTM networks in a variety of configurations in terms of number of units and layers. In all cases, the performance of LSTMs was not competitive relative to other models, so we do not report the results.

[Figure 2 around here]

With the exception of the HAR and HARX models, the ML methods we use require selection or tuning of one or more hyperparameters. For most methods, the training window at each iteration is used to (i) select the combination of hyperparameters with the best performance using 10-fold cross-validation, and (ii) to fit the optimal model using the selected hyperparameters. The exception are the NNs, for which we apply early stopping, using the last 252 daily observations as a validation sample to determine the optimal number of epochs to stop the training. We also rely on an ensemble approach to reduce any dependency on the starting weights of the networks. This is done for each NN topology by repeating the training (with early stopping) 50 times and averaging the predictions from the 50 NNs.

For linear, Bagging and Random Forest methods, computations were performed using R 3.6.3 package (R Core Team, 2020). Specifically, HAR and HARX estimates were obtained using the `lm` package. Lasso, Ridge and Elastic Net were fitted using the `glmnet` package, with the hyperparameter λ as the default value. Bagging and Random Forest methods were fitted using the `randomForest` package. For both methods, the hyperparameter `ntree` was tested on the grid 100, 200, ..., 1000. In addition, the hyperparameter `mtry`, which controls the number of variables to be sub-sampled, was set as the total number of predictors for Bagging, whereas for Random Forests, values ranging from the floor of the squared root of the number of predictors to half the number of predictors were tested. The remaining hyperparameters were set to default values. For neural network methods, the computations were performed in Matlab 2020b using the Deep Learning toolbox.

4.3 Assessing Out-Of-Sample Performance

Let $t = S + 1, \dots, T$ denote the out-of-sample (OOS) testing period. We consider two metrics to evaluate the out-of-sample forecasting performance:

1. **Statistical accuracy.** Based on the root mean square error (RMSE). Specifically, we computed the estimated k -step ahead RMSE as

$$RMSE_k = \left\{ \frac{1}{T - k - S} \sum_{t=S+1}^{T-k} (\sigma_{t+k} - \hat{\sigma}_{t+k/t})^2 \right\}^{1/2}. \quad (8)$$

2. **One-day-ahead Value at Risk (VaR) forecasting performance.** Some studies suggest that the RV may be useful to forecast daily VaR (Giot and Laurent, 2004; Louzis et al., 2014; Wong et al., 2016). Considering the important role of VaR for practitioners and regulators, we build 1-day-ahead VaR forecasts based on the corresponding RV forecasts from different models. The procedure we use to forecast VaR is as follows. We assume that returns are generated through $r_t = \mu + \sigma_t \varepsilon_t$, where $\{\varepsilon_t\}$ is an iid sequence with zero mean and variance equal to one, and ε_t is independent of σ_t for all t . To estimate the VaR of confidence $(1 - \alpha)$ at time $t + 1$, based on the information up to time t , denoted by $VaR_{t+1/t}^\alpha$, a two-step procedure is employed. First, we calculate $\hat{\mu}$ and $\hat{\sigma}_{t+1/t}$ by some method and then we compute $VaR_{t+1/t}$ as

$$VaR_{t+1/t}^\alpha = \hat{\mu} + q_\alpha \hat{\sigma}_{t+1/t}, \quad (9)$$

where q_α is the α -quantile of the distribution of ε_t . Rather than assuming a specific parametric distribution for ε_t , at each point in time we use the actual values of RV in the training set (either *rv5* or *rvts*) to calculate the standardized innovations, $(r_t - \bar{r})/\sigma_t$, and then estimate q_α as its α -quantile. $\hat{\mu}$ is calculated as the mean return in each training dataset ($\bar{r}$), and $\hat{\sigma}_{t+1/t}$ is taken to be the 1-day-ahead forecast of RV based on each ML model.

To assess the performance of each method in terms of the VaR, we compare the estimated $VaR_{t+1/t}(\alpha)$ with the observed return r_{t+1} for each time in the out-of-sample period, and compute the percentage of violations (exceptions), i.e. the percentage of values such that r_{t+1} is smaller than its estimated VaR. In addition, we applied the Kupiec (1995) test to check if the proportion of violations was close to the theoretical value, the Christoffersen (1998) test to

check conditional coverage, and the [Christoffersen and Pelletier \(2004\)](#) test to check if the violations were independent.

4.3.1 Model Comparison

We compare the forecasting performance of different models using the model confidence set (MCS) approach of [Hansen et al. \(2011\)](#). Given a set $\mathcal{M}$ of models, the MCS approach is based on a sequence of pairwise significance tests, in which models which are found to be significantly inferior are eliminated. When no further elimination is possible, the remaining models form the set of superior models (SSM), which is interpreted to contain the best model with some confidence. More formally, assume there are m models in $\mathcal{M}$, and let $L_{i,t} = L(\sigma_t, \hat{\sigma}_t)$ denote a loss function for model i at time t , where σ_t and $\hat{\sigma}_t$ are the observed and predicted RV at time t , respectively. The relative performance between two models i and j is defined as $d_{ij,t} = L_{i,t} - L_{j,t}$. The hypotheses tested in the MCS procedure are of the form $H_{0,\mathcal{M}} : E(d_{ij,t}) = 0, \quad \forall i, j \in \mathcal{M}$. The test statistic has a nonstandard distribution, and p -values are obtained using bootstrap techniques. All tests are performed using the MCS R package with a confidence level of 0.20, see [Bernardi and Catania \(2018\)](#) for details.¹¹

When comparing models in terms of their ability to forecast RV, we consider the quadratic loss function

$$L_{i,t} = (\sigma_t - \hat{\sigma}_t)^2. \quad (10)$$

For comparison of VaR forecasts from different models, we use the asymmetric loss function of [González-Rivera et al. \(2004\)](#):

$$L_{i,t}(r_t, VaR_t) = (\alpha - d_t^\alpha)(r_t - VaR_t^\alpha), \quad (11)$$

where VaR_t^α is the $(1-\alpha)\%$ VaR forecast for date t , and $d_t^\alpha = \mathbb{1}(r_t < VaR_t^\alpha)$. This function penalizes more heavily (i.e., with weight $(1 - \alpha)$) instances in which the

¹¹Based on the recommendation in the corrigendum to [Hansen et al. \(2011\)](#), we use the TR statistic to test the hypothesis.

observed return is lower than the VaR estimate. This approach is similar to the one followed by [Bernardi and Catania \(2016\)](#).

5 Empirical Results

5.1 Descriptive Statistics

Figure 1 shows the daily returns and *rvts* estimates of RV for each of the ten indices.¹² The graphs show that the RV estimates react immediately to periods of stress in the market, with peaks around important episodes of increased market turmoil, such as the 2008 crisis, the European sovereign debt crisis from 2009 to 2012, and the recent COVID-19 pandemic-induced market panic in early 2020. Table 3 reports descriptive statistics of daily returns and the two RV measures for each index. The table reveals some interesting patterns. First, daily index returns have averages close to zero, are negatively skewed and leptokurtic. Second, for all indices, the average *rv5* and *rvts* realized volatilities are lower than the daily standard deviations of returns. The daily standard deviation of the returns of the indices ranges from 1.162% for the FTSE index (approximately 18% on an annual basis) to 1.805% for the BVSP index ($\sim 29\%$ on an annual basis), while the mean *rv5* measure, for example, ranges from 0.842% for the S&P500 index ($\sim 13\%$ on an annual basis) to 1.096% for the BVSP index ($\sim 17\%$ on an annual basis). This can be explained by the fact that the daily returns measure variation from one day's closing price to the next, and therefore includes information from overnight events (see, e.g., [Tsiakas, 2008](#); [Ahoniemi and Lanne, 2013](#)).

Interestingly, the indices with the highest daily standard deviations of returns are not necessarily those with the highest mean RV measures, although this is true for the Brazilian index (BVSP), which has the highest daily standard deviation, at around 1.805% per day, and the highest mean *rv5* and *rvts*, at 1.096% and 1.056%, respectively). For example, the second most volatile index in terms of daily standard

¹²The *rv5* estimates are quite similar to the *rvts* ones and are omitted from the graph to reduce clutter.

deviation is the Nasdaq index (daily standard deviation of 1.577%), which is only the sixth most volatile in terms of the mean *rv5* (mean of 0.924%) and the seventh in terms of the mean *rvts* (mean of 0.837%).

We also note that the two measures of RV are extremely skewed to the right and leptokurtic.

[Figure 1 around here]

[Table 3 around here]

5.2 Statistical Accuracy of RV Forecasts

Tables 4 and 5 report out-of-sample RMSE values, calculated by (8), for 1-day-ahead volatility forecasts using *rv5* and *rvts* measures, respectively. Within each table, for each index, the model (or models) with the lowest RMSE are shown in bold, and models in the SSM are shown in shaded color.

[Tables 4 and 5 around here]

We start by discussing the results obtained with the *rv5* measure in Table 4. The first important conclusion from this table is that the additional predictors are helpful in forecasting 1-day-ahead *rv5*. For all indices, the RMSE of the HARX model, which includes the additional predictors, is lower than the RMSE of the HAR model, which only includes the lagged moving averages of RV. The differences appear to be statistically significant: the HAR model is not included in the SSM for 8 out of 10 indices, suggesting that models with additional predictors dominate HAR in terms of having lower forecasting errors.

When we consider only linear models with all available predictors (i.e., HARX, Lasso, Ridge, and Enet), the differences in RMSE are not dramatic. With a few exceptions (the Ridge model for the EURO50 and FTSE100 indices; and the Enet model for the Euro50 and Nasdaq100 indices), linear models with all predictors are always included in the SSM.

A natural question that we seek to answer is whether more sophisticated models, which allow for nonlinear relationships between RV and the predictors, outperform

the simpler linear models. If we focus on the best performing model in terms of lowest RMSE, we observe that linear models were the best performers four times (the HARX model once for the BVSP index, and Lasso three times for the FTSE100, Nasdaq100, and Nikkei225 indices), while nonlinear models were the best performers six times (the RF method was the best performer for the CAC40, DAX, EURO50, and NIFTY50 indices; NNs were the best performers for the SCI and S&P500 indices). However, when we consider the SSM, the differences are not statistically significant: for all indices, linear models are always included in the SSM, even when the best performer in terms of RMSE is a nonlinear model. These results show that there is no clear dominance of nonlinear relative to linear models (and vice-versa).

The results using the *rvts* measure, shown in Table 5, show similarities with those for the *rv5* measure. First, including additional predictors improves performance. For all indices, the RMSE of the HARX model is lower than that of the HAR. Second, as with the *rv5* measure, there is no dominance of nonlinear models relative to linear models: the lowest RMSE is obtained for linear models for 5 out of 10 indices (HARX for the BVSP and Nasdaq100 indices; Lasso for the FTSE100, Nikkei225, and SCI indices), with nonlinear models showing the lowest RMSE for the remaining 5 indices (the RF model for the DAX and NIFTY50 indices; Boosting for the EURO50 index, and neural networks for the CAC40 and S&P500 indices). Third, even in the cases in which a nonlinear model achieved the lowest RMSE, the SSM always includes linear models, suggesting that the differences in performance are not statistically significant.

Overall, these results suggest that there is not much benefit in using complex models, such as the nonlinear models considered in this study, to forecast 1-day-ahead RV. Even when a nonlinear model has the best performance in terms of RMSE, there is always at least one linear model in the SSM. In particular, the HARX or its Lasso version are always included in the SSM for all indices.

Interestingly, when we analyze results with longer forecasting horizon, the evidence turns even more strongly in favor of using linear models. Table 6 shows results for a forecasting horizon of five days for *rvts* (similar results are obtained for *rv5* and are

shown on Table A1 in the Appendix. The HAR model has the lowest RMSE and is always included in the SSM for all stock indices. Other linear methods are usually also included in the SSM. In contrast, nonlinear models are almost always excluded from the SSM. Similar conclusions are obtained for a 22-day forecast horizon (Tables A2 and A3). We draw two conclusions from these results. First, the benefit of including additional variables only seems to exist for very near-term (i.e., 1-day-ahead) RV forecasting. Second, simple linear models of past RV provide the best forecasting performance when the forecasting horizon increases.

[Table 6 around here]

5.3 VaR Forecasting

We forecast 1-day-ahead VaR using RV forecasts from different models, following the methodology discussed in Section 4.3. For brevity, we focus on VaR forecasts obtained using *rvts* RV measure. Tables 7 and 8 report the frequency of violations obtained for each index for VaR levels of 5% and 1%, respectively. The models with a frequency of VaR violation closest to the VaR α level are shown in bold, while models in the SSM, considering the loss function (10), are shown in shaded color. In addition to the frequency of VaR violations, we report in the Appendix for both the *rvts* and the *rv5* measures: the p-value for the Kupiec (1995) test, which compares the percentage of violations with its theoretical value (i.e., α), the p-value for Christoffersen (1998) conditional coverage test, and the p-value for the Christoffersen and Pelletier (2004) test (*CP*), which tests whether VaR violations are independent.

We start by discussing the results for the 5% VaR (i.e., $1 - \alpha = 0.95$) in Table 7. A few noteworthy features emerge. First, we observe a general tendency of underestimation of the VaR, i.e., for a given index, the percentage of violations is typically above the VaR level α . Indeed, the results of the Kupiec (1995) test (Tables A4 to A13 in the Appendix) suggest that, with a few exceptions (the Ridge model for the Nasdaq100 index and all models for the SCI index), the hypothesis that the frequency of violations is equal to 5% is rejected. Second, the HAR and HARX models are out-

side the SSM most of the time (six and seven times, among ten, respectively). Third, the Ridge method provides the closest frequency figures to the nominal confidence value (eight times) and this method is present nine times in the SSM. In this sense, this method can be considered the best. The second best method is RF, which is also present nine times in the SSM. Finally, in those situations when nonlinear ML methods are included in the SSM, at least one linear method is always included.

Similar results are obtained for 1% VaR in Table 8. For instance, there exists VaR underestimation and HAR and HARX are outside the SSM most of the time. Furthermore, the best methods are Ridge and RF (each is included eight times in the SSM and both present the best frequency violation numbers). Again, linear models are also always present in the SSM, except for the NIFTY 50 index.

Figure 3 presents the estimated 1-day-ahead 1%-VaRs with *rvts* for the BVSP and S&P500 indices using some selected models. For each series, this picture reveals very good behavior for the three models, in the sense that the VaR values mimic very well the movements of the volatility and they are not far below the real losses. Moreover, in both series we observe that the VaR estimates provided by the methods are usually very close, especially the linear ones. However, in periods of very high volatility, such as the middle of 2008, the forecasts provided by the nonlinear methods are decoupled from the others. Although not shown in the graph, similar behavior was observed for the other indices.

[Figure 3 around here]

6 Conclusions

In this paper, we evaluate the potential benefits of machine learning methods in the task of forecasting different realized volatility (RV) measures for ten global stock market indices. We conduct an extensive out-of-sample forecasting exercise with different linear and nonlinear machine learning methods, using as predictors lags of RV, as well as additional macroeconomic variables. We compare models using the model confidence set approach of Hansen et al. (2011), which identifies a set of

superior models (SSM) containing the best model for a given confidence level. We also assess the performance of Value-at-Risk (VaR) forecasts based on volatility forecasts from different models.

Our main results can be summarized as follows. First, we show that additional predictors improve forecasting performance of 1-day-ahead RV for all indices, relative to the Heterogeneous Autoregressive (HAR) model of [Corsi \(2009\)](#). Indeed, for all stock indices, the out-of-sample Root Mean Squared Error (RMSE) of the HAR model is always higher than that of models that include all available predictors. In addition, the HAR model is typically not included in the SSM. On the other hand, including additional variables does not seem helpful to forecast 5 or 22-day-ahead RV. Second, we find that nonlinear machine learning methods, such as Bagging, Random Forests, Boosting, and neural networks, in general do not outperform simpler linear models. In a few cases, particularly for 1-day-ahead RV forecasting, nonlinear models are among the models with lowest RMSE. However, even in those cases, linear models using all available predictors, with or without regularization, are almost always among the top performers, and one or more linear models are always included in the SSM. Finally, we compare the RV forecasting methods in terms of VaR estimation. We find that, most of the time, the Ridge model was the best performer in terms of the frequency of VaR violations. However, regardless of the RV measure and stock index, we found a general tendency to underestimate VaR. We also find that VaR forecasts based on the realized kernel variance (two-scale/Bartlett) RV measure of [Barndorff-Nielsen et al. \(2008\)](#) fare better than VaR forecasts using the 5-minute RV measure.

Our study has implications for Finance practitioners as well as academics. Our results suggest that practitioners may rely on linear models (possibly with regularization) to forecast daily RV. These models are simpler, computationally inexpensive, and readily interpretable. For academics, our results shed light on the determinants of RV, showing that lagged returns and exogenous macroeconomic variables are useful predictors of RV, but only at short (i.e., 1-day) forecast horizons.

Acknowledgements

Rafael R. Branco and Mauricio Zevallos acknowledge support of the Centre for Applied Research on Econometrics, Finance and Statistics (CAREFS). Mauricio Zevallos acknowledges the financial support of FAPESP 2018/04654-9 grant.

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Table 1: Stock market indices

Index		Country/Region	Period
BVSP BOVESPA Index	BVSP	Brazil	Jan 04, 2000 to Dec 30, 2021
CAC 40	CAC40	France	Jan 04, 2000 to Dec 31, 2021
DAX	DAX	Germany	Jan 04, 2000 to Dec 30, 2021
EURO STOXX 50	EURO50	Europe	Jan 04, 2000 to Dec 31, 2021
FTSE 100	FTSE	UK	Jan 05, 2000 to Dec 31, 2021
Nasdaq 100	Nasdaq	US	Jan 04, 2000 to Dec 31, 2021
NIFTY 50	NIFTY	India	Jan 04, 2000 to Dec 31, 2021
Nikkei 225	Nikkei	Jappan	Feb 03, 2000 to Dec 30, 2021
Shanghai Composite Index	SCI	China	Jan 05, 2000 to Dec 31, 2021
S&P 500 Index	S&P500	US	Jan 04, 2000 to Dec 31, 2021

Table 2: List of predictors used in the study

Variable	Definition	Source
$RV_t^{(n)}$	Moving averages of RV for $n = 1, 5$, and 22 days	Realized Library
$r_t^{(n)}$	Moving averages of past returns for $n = 1, 5$, and 22 days	Realized Library
$[r_t^{(n)}]^2$	Square of $r_t^{(n)}$ for $n = 1, 5$, and 22 days	Realized Library
ΔSTR_t	Daily change in short-term benchmark interest rate	Refinitiv
ΔLTR_t	Daily change in long-term benchmark interest rate	Refinitiv
ΔDP_t	Daily index dividend yield	Refinitiv
ΔEP_t	Daily index earnings-to-price ratio	Refinitiv
FX_t	Daily return of country exchange rate	Refinitiv
$Comm_t$	Daily return of the S&P GSCI Commodity Index	Refinitiv
ΔVIX_t	Daily change in VIX index	Refinitiv

Table 3: Descriptive statistics of daily returns and realized volatilities (*rv5* and *rvts*).

Index	<i>n</i>	Daily returns				<i>rv5</i> realized volatility				<i>rvts</i> realized volatility			
		Mean ($\times 100$)	Std. Dev. ($\times 100$)	Skew	Kurt	Mean ($\times 100$)	Std. Dev. ($\times 100$)	Skew	Kurt	Mean ($\times 100$)	Std. Dev. ($\times 100$)	Skew	Kurt
BVSP	5408	0.034	1.805	-0.308	9.737	1.096	0.585	3.518	23.851	1.056	0.528	3.541	24.507
CAC40	5611	0.003	1.408	-0.239	8.955	0.988	0.588	2.835	18.095	0.979	0.578	2.699	16.086
DAX	5573	0.015	1.453	-0.168	9.431	1.062	0.672	2.625	14.716	1.040	0.651	2.767	16.773
EURO50	5595	-0.002	1.426	-0.214	8.798	1.055	0.670	3.221	23.231	1.023	0.619	2.817	17.536
FTSE	5546	0.002	1.162	-0.298	10.234	0.917	0.603	3.833	32.332	0.809	0.533	3.931	35.272
Nasdaq	5515	0.024	1.577	-0.178	9.773	0.924	0.616	2.710	15.855	0.837	0.581	2.660	15.418
Nikkei	5346	0.007	1.478	-0.390	9.263	0.876	0.489	2.914	19.407	0.857	0.463	2.766	17.708
NIFTY	5454	0.044	1.445	-0.492	13.102	0.919	0.644	5.154	62.741	0.855	0.606	4.183	39.528
SCI	5313	0.018	1.524	-0.373	8.155	1.057	0.664	2.322	11.564	0.689	0.404	2.980	19.626
S&P500	5514	0.022	1.232	-0.393	13.741	0.842	0.612	3.375	22.837	0.787	0.559	3.566	26.469

n is the total number of daily observations. The sample means (Mean) and standard deviations (Std. Dev.) are multiplied by 100.

Table 4: RMSE values ($\times 1000$) of 1-day-ahead $rv5$ volatility forecasts. Models in the Superior Set of Models of Hansen et al. (2011) for each index are shown in shaded color. Bold values correspond to the lowest RMSE.

Method/Index	BVSP	CAC40	DAX	EURO50	FTSE100	Nasdaq100	NIFTY50	Nikkei225	SCI	S&P500
HAR	3.615	3.166	3.206	3.931	3.898	2.950	4.235	3.211	4.003	3.475
HARX	3.486	2.987	3.059	3.753	3.727	2.789	3.985	3.042	3.763	3.265
Lasso	3.499	2.982	3.054	3.744	3.725	2.785	4.065	3.024	3.738	3.273
Ridge	3.509	2.995	3.080	3.771	3.751	2.795	3.991	3.047	3.751	3.313
Enet	3.500	2.986	3.055	3.748	3.727	2.789	4.058	3.029	3.747	3.280
Bagging	4.310	3.048	3.099	3.766	3.900	2.852	4.121	3.301	3.935	3.730
RF	4.235	2.980	3.045	3.709	3.755	2.818	3.732	3.264	3.887	3.676
Boosting	4.308	3.026	3.045	3.731	3.823	2.829	3.916	3.205	3.934	3.665
NN1	3.986	3.097	3.173	4.160	4.260	2.983	4.526	3.120	3.826	3.337
NN2	4.080	3.070	3.198	3.925	3.903	3.045	4.208	3.067	3.736	3.282
NN3	3.937	2.926	3.099	3.808	3.776	2.923	4.176	3.070	3.719	3.237
NN4	4.001	2.932	3.066	3.818	3.735	2.901	3.956	3.136	3.879	3.230
NN5	4.052	2.976	3.064	3.751	3.708	2.906	4.045	3.204	3.881	3.349

Table 5: RMSE values ($\times 1000$) of 1-day-ahead *rvts* volatility forecasts. Models in the Superior Set of Models of Hansen et al. (2011) for each index are shown in shaded color. Bold values correspond to the lowest RMSE.

Method/Index	BVSP	CAC40	DAX	EURO50	FTSE100	Nasdaq100	NIFTY50	Nikkei225	SCI	S&P500
HAR	3.204	2.906	2.781	3.301	3.023	2.499	3.903	2.813	2.258	2.862
HARX	3.072	2.771	2.694	3.170	2.924	2.378	3.651	2.657	2.092	2.691
Lasso	3.085	2.768	2.685	3.164	2.921	2.381	3.649	2.642	2.088	2.704
Ridge	3.085	2.786	2.706	3.186	2.954	2.393	3.641	2.675	2.106	2.741
Enet	3.082	2.769	2.680	3.161	2.923	2.386	3.658	2.646	2.089	2.707
Bagging	3.820	2.822	2.760	3.137	3.241	2.458	3.798	2.896	2.395	3.273
RF	3.773	2.766	2.671	3.102	3.030	2.406	3.534	2.894	2.428	3.260
Boosting	3.780	2.803	2.771	3.101	3.083	2.426	3.648	2.834	2.476	3.192
NN1	3.539	2.901	3.008	3.278	3.329	2.559	3.783	2.730	2.204	2.843
NN2	3.724	2.806	2.852	3.354	3.093	2.541	3.763	2.716	2.143	2.698
NN3	3.572	2.767	2.822	3.227	2.970	2.498	3.666	2.716	2.136	2.680
NN4	3.589	2.709	2.787	3.133	2.933	2.438	3.695	2.739	2.161	2.705
NN5	3.731	2.730	2.751	3.106	2.989	2.498	3.707	2.814	2.218	2.871

Table 6: RMSE values ($\times 1000$) of 5-day-ahead *rvts* volatility forecasts. Models in the Superior Set of Models of Hansen et al. (2011) for each index are shown in shaded color. Bold values correspond to the lowest RMSE.

Method/Index	BVSP	CAC40	DAX	EURO50	FTSE100	Nasdaq100	NIFTY50	Nikkei225	SCI	S&P500
HAR	3.210	2.908	2.783	3.302	3.025	2.500	3.915	2.814	2.260	2.863
HARX	3.239	2.935	2.789	3.306	3.042	2.515	3.943	2.927	2.281	2.890
Lasso	3.257	2.928	2.795	3.305	3.029	2.502	3.924	2.851	2.277	2.880
Ridge	3.275	2.950	2.805	3.324	3.060	2.546	3.925	2.836	2.298	2.930
Enet	3.267	2.923	2.793	3.298	3.035	2.507	3.922	2.843	2.279	2.884
Bagging	3.776	3.086	2.932	3.444	3.230	2.652	4.132	3.217	2.692	3.343
RF	3.849	3.033	2.953	3.458	3.229	2.579	3.975	3.187	2.684	3.533
Boosting	4.795	4.399	4.177	4.847	4.221	3.846	5.247	4.112	3.363	4.764
NN1	5.112	4.524	4.479	5.130	4.453	3.784	5.150	3.956	3.509	4.779
NN2	5.112	4.385	4.450	4.888	4.185	3.756	4.996	3.903	3.465	4.685
NN3	4.948	4.321	4.288	4.806	4.112	3.755	5.009	3.918	3.394	4.583
NN4	4.978	4.332	4.272	4.732	4.095	3.755	4.980	3.958	3.366	4.606
NN5	4.948	4.331	4.208	4.769	4.227	3.752	5.058	4.033	3.431	4.705

Table 7: Frequency of 5% VaR violations based on *rvts* volatility forecasts. Models in the Superior Set of Models of Hansen et al. (2011) for each index are shown in shaded color. Bold values correspond to frequencies of VaR violations closest to the VaR level.

Method/Index	BVSP	CAC40	DAX	EURO50	FTSE100	Nasdaq100	NIFTY50	Nikkei225	SCI	S&P500
HAR	0.062	0.073	0.092	0.077	0.071	0.060	0.034	0.086	0.054	0.074
HARX	0.060	0.069	0.092	0.070	0.068	0.062	0.031	0.074	0.056	0.074
Lasso	0.059	0.068	0.090	0.070	0.067	0.061	0.029	0.073	0.053	0.073
Ridge	0.059	0.067	0.085	0.068	0.066	0.057	0.028	0.071	0.053	0.071
Enet	0.059	0.069	0.090	0.070	0.067	0.059	0.029	0.073	0.054	0.072
Bagging	0.060	0.069	0.088	0.071	0.068	0.061	0.032	0.076	0.052	0.072
RF	0.060	0.069	0.086	0.071	0.067	0.061	0.031	0.077	0.055	0.071
Boosting	0.062	0.071	0.086	0.070	0.071	0.060	0.035	0.076	0.058	0.075
NN1	0.064	0.069	0.090	0.069	0.072	0.064	0.034	0.078	0.056	0.075
NN2	0.063	0.069	0.090	0.069	0.071	0.064	0.034	0.075	0.056	0.075
NN3	0.062	0.069	0.090	0.068	0.070	0.062	0.032	0.075	0.056	0.075
NN4	0.061	0.070	0.090	0.070	0.069	0.063	0.034	0.073	0.056	0.074
NN5	0.061	0.069	0.090	0.069	0.070	0.063	0.034	0.074	0.056	0.074

Table 8: Frequency of 1% VaR violations based on *rvts* volatility forecasts. Models in the Superior Set of Models of Hansen et al. (2011) for each index are shown in shaded color. Bold values correspond to frequencies of VaR violations closest to the VaR level.

Method/Index	BVSP	CAC40	DAX	EURO50	FTSE100	Nasdaq100	NIFTY50	Nikkei225	SCI	S&P500
HAR	0.014	0.0301	0.046	0.033	0.034	0.029	0.003	0.041	0.027	0.040
HARX	0.013	0.029	0.044	0.031	0.030	0.030	0.002	0.032	0.027	0.042
Lasso	0.013	0.029	0.042	0.031	0.029	0.029	0.002	0.030	0.027	0.040
Ridge	0.013	0.028	0.041	0.030	0.029	0.028	0.002	0.030	0.026	0.039
Enet	0.013	0.029	0.042	0.031	0.029	0.029	0.002	0.031	0.027	0.039
Bagging	0.014	0.028	0.042	0.031	0.032	0.030	0.002	0.034	0.026	0.040
RF	0.015	0.028	0.042	0.029	0.031	0.028	0.002	0.035	0.026	0.039
Boosting	0.015	0.030	0.043	0.031	0.031	0.027	0.002	0.037	0.030	0.041
NN1	0.016	0.031	0.043	0.032	0.033	0.031	0.002	0.034	0.029	0.041
NN2	0.015	0.029	0.045	0.032	0.031	0.032	0.002	0.033	0.028	0.041
NN3	0.015	0.029	0.045	0.032	0.031	0.031	0.002	0.032	0.028	0.040
NN4	0.015	0.029	0.045	0.031	0.031	0.032	0.002	0.034	0.029	0.040
NN5	0.014	0.030	0.045	0.032	0.032	0.031	0.002	0.034	0.029	0.040

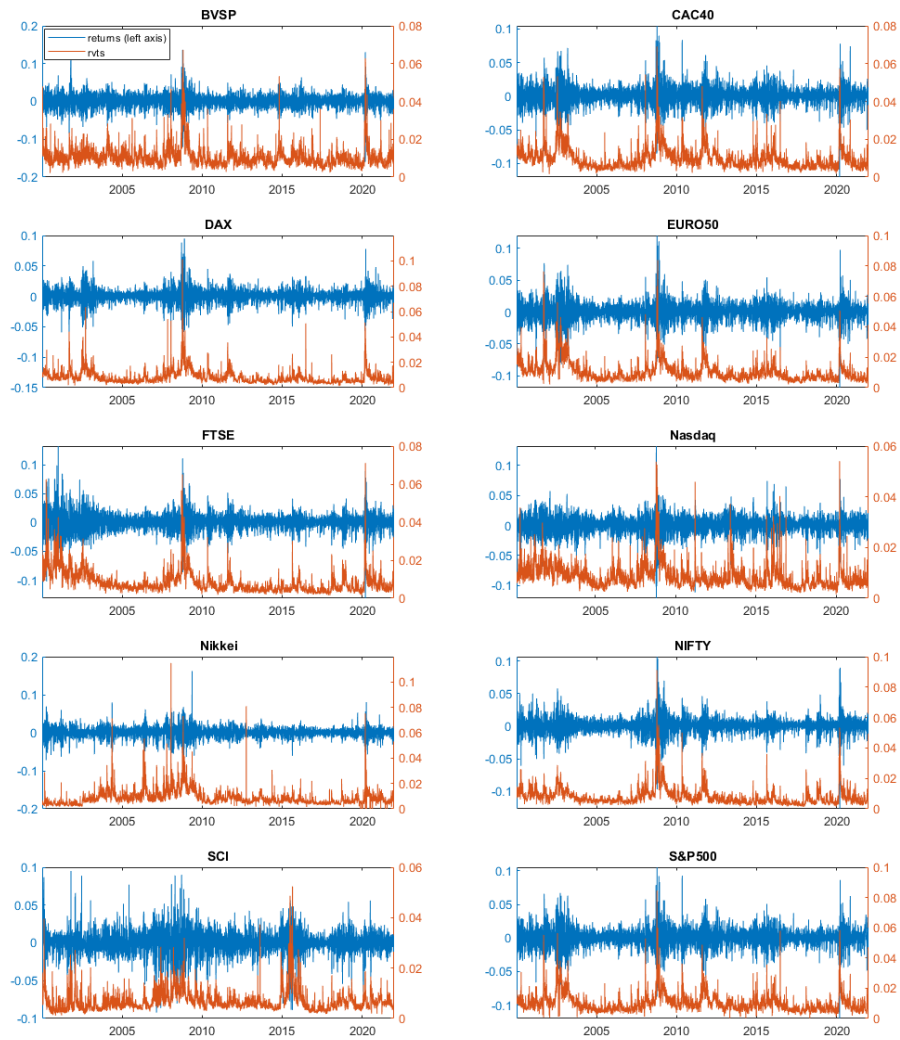


Figure 1: Daily returns and realized volatility estimates for ten stock indices

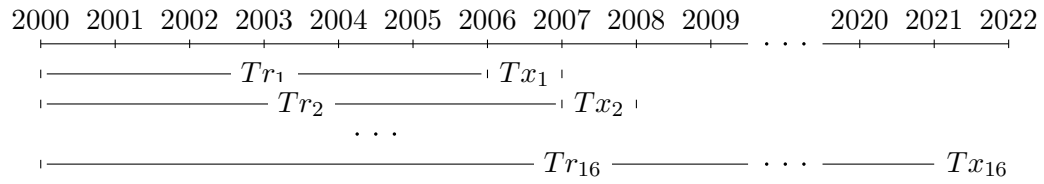


Figure 2: Graphical representation of the training and testing sample splitting approach

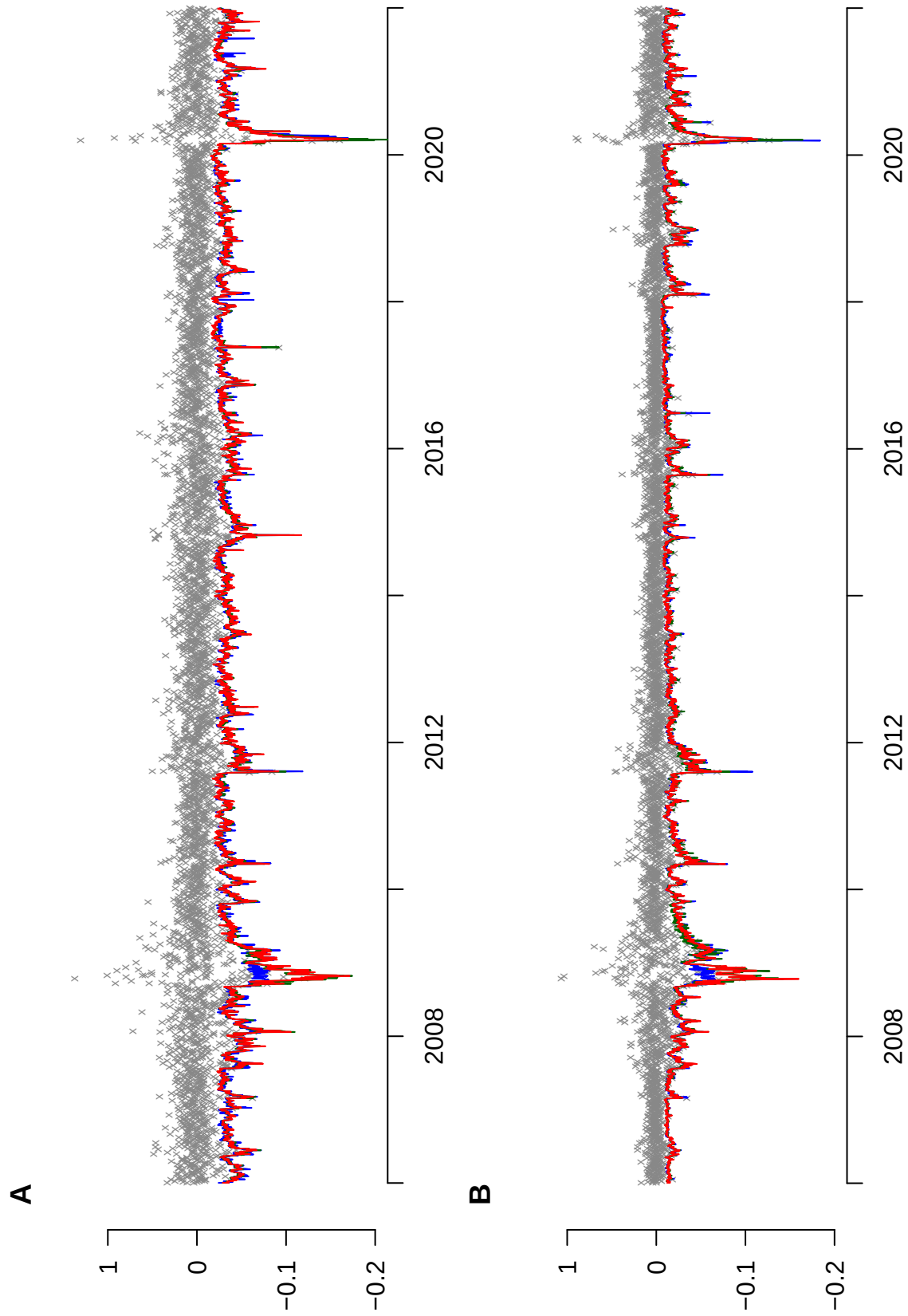


Figure 3: 1-day-ahead 1% VaR. (A) BVSP: HAR (in red), Lasso (in green) and Bagging (in blue). (B): S&P 500. HAR (in red), Enet (in green) and Bagging (in blue).

A Appendix

Table A1: RMSE values ($\times 1000$) of 5-day-ahead $rv5$ volatility forecasts. Models in the Superior Set of Models of Hansen et al. (2011) for each index are shown in shaded color. Bold values correspond to the lowest RMSE.

Method/Index	BVSP	CAC40	DAX	EURO50	FTSE100	Nasdaq100	NIFTY50	Nikkei225	SCI	S&P500
HAR	3.620	3.168	3.208	3.932	3.900	2.951	4.245	3.212	4.005	3.476
HARX	3.660	3.199	3.206	3.942	3.914	2.970	4.276	3.296	4.052	3.512
Lasso	3.674	3.188	3.210	3.934	3.897	2.959	4.264	3.229	4.036	3.495
Ridge	3.698	3.208	3.225	3.957	3.920	2.999	4.272	3.224	4.055	3.538
Enet	3.703	3.186	3.209	3.932	3.897	2.960	4.261	3.222	4.035	3.499
Bagging	4.310	3.350	3.327	4.156	4.073	3.146	4.406	3.638	4.390	3.979
RF	4.332	3.287	3.353	4.130	4.080	3.063	4.308	3.577	4.300	4.084
Boosting	5.483	4.559	4.554	5.462	5.006	4.359	5.874	4.441	5.579	5.356
NN1	5.722	4.708	4.74	5.903	5.210	4.408	5.732	4.241	5.843	5.401
NN2	5.725	4.596	4.703	5.617	5.038	4.326	5.543	4.230	5.665	5.333
NN3	5.599	4.529	4.573	5.413	4.919	4.317	5.543	4.271	5.691	5.204
NN4	5.405	4.549	4.566	5.403	4.953	4.334	5.517	4.295	5.525	5.202
NN5	5.511	4.513	4.537	5.414	4.974	4.307	5.63	4.298	5.582	5.285

Table A2: RMSE values ($\times 1000$) of 22-day-ahead $rv5$ volatility forecasts. Models in the Superior Set of Models of Hansen et al. (2011) for each index are shown in shaded color. Bold values correspond to the lowest RMSE.

Method/Index	BVSP	CAC40	DAX	EURO50	FTSE100	Nasdaq100	NIFTY50	Nikkei225	SCI	S&P500
HAR	3.603	3.177	3.205	3.940	3.909	2.956	4.101	3.204	4.01	3.481
HARX	3.614	3.183	3.211	3.953	3.921	2.971	4.112	3.216	4.043	3.520
Lasso	3.630	3.178	3.208	3.940	3.912	2.960	4.120	3.214	4.026	3.503
Ridge	3.639	3.194	3.222	3.961	3.931	2.989	4.138	3.219	4.051	3.538
Enet	3.629	3.179	3.207	3.941	3.915	2.957	4.152	3.213	4.025	3.508
Bagging	3.983	3.334	3.339	4.084	4.173	3.128	4.500	3.591	4.426	3.869
RF	4.123	3.336	3.346	4.147	4.165	3.103	4.205	3.550	4.287	4.047
Boosting	6.405	5.732	5.899	6.605	6.106	5.457	6.350	5.284	6.441	6.573
NN1	6.372	5.559	5.927	6.694	6.034	5.461	6.144	5.205	6.510	6.530
NN2	6.305	5.510	5.766	6.524	5.970	5.401	6.012	5.105	6.416	6.413
NN3	6.221	5.485	5.683	6.41	5.927	5.354	6.019	5.100	6.390	6.350
NN4	6.297	5.534	5.697	6.393	5.981	5.356	6.048	5.141	6.301	6.366
NN5	6.454	5.607	5.761	6.454	6.118	5.310	6.129	5.135	6.242	6.414

Table A3: RMSE values ($\times 1000$) of 22-day-ahead *rvts* volatility forecasts. Models in the Superior Set of Models of Hansen et al. (2011) for each index are shown in shaded color. Bold values correspond to the lowest RMSE.

Method/Index	BVSP	CAC40	DAX	EURO50	FTSE100	Nasdaq100	NIFTY50	Nikkei225	SCI	S&P500
HAR	3.190	2.913	2.773	3.308	3.030	2.504	3.640	2.806	2.264	2.869
HARX	3.200	2.919	2.782	3.318	3.042	2.516	3.648	2.818	2.282	2.899
Lasso	3.209	2.915	2.778	3.308	3.035	2.506	3.649	2.811	2.275	2.881
Ridge	3.227	2.933	2.798	3.329	3.056	2.538	3.658	2.828	2.299	2.926
Enet	3.214	2.915	2.776	3.308	3.038	2.509	3.652	2.814	2.278	2.890
Bagging	3.501	3.090	2.906	3.433	3.286	2.652	4.116	3.206	2.609	3.330
RF	3.704	3.064	2.936	3.459	3.285	2.628	3.829	3.185	2.658	3.535
Boosting	5.960	5.598	5.615	6.010	5.285	4.881	6.041	4.991	4.069	6.145
NN1	5.899	5.393	5.616	6.058	5.167	4.841	5.869	4.876	4.001	5.910
NN2	5.836	5.315	5.475	5.873	5.103	4.792	5.759	4.829	3.957	5.787
NN3	5.743	5.279	5.415	5.777	5.029	4.762	5.728	4.824	3.911	5.783
NN4	5.827	5.361	5.419	5.773	5.141	4.773	5.790	4.848	3.936	5.807
NN5	5.867	5.411	5.464	5.813	5.261	4.732	5.854	4.865	4.092	5.871

Table A4: Backtesting results of 1-day-ahead VaR for the BVSP Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	RV5				RVTS			
		<i>freq</i>	K	C	CP	<i>freq</i>	K	C	CP
0.05	HAR	0.062	0.001	0.001	0.83	0.056	0.096	0.097	0.909
	HARX	0.060	0.007	0	0.293	0.056	0.096	0.012	0.615
	Lasso	0.059	0.008	0	0.364	0.054	0.272	0.016	0.415
	Ridge	0.059	0.008	0	0.405	0.055	0.189	0.028	0.711
	Enet	0.059	0.010	0.001	0.330	0.054	0.242	0.014	0.422
	Bagging	0.060	0.007	0.001	0.170	0.053	0.339	0.059	0.954
	RF	0.060	0.005	0	0.215	0.055	0.146	0.020	0.953
	Boosting	0.062	0.001	0	0.370	0.056	0.096	0.004	0.701
	NN1	0.064	0	0	0.312	0.057	0.044	0.011	0.910
	NN2	0.063	0	0	0.430	0.058	0.022	0.010	0.949
	NN3	0.062	0.001	0	0.470	0.056	0.096	0.028	0.917
	NN4	0.061	0.002	0	0.503	0.057	0.051	0.013	0.960
	NN5	0.061	0.002	0	0.813	0.057	0.044	0.022	0.522
0.01	HAR	0.014	0.008	0.029	0.797	0.012	0.300	0.497	0.678
	HARX	0.013	0.075	0.104	0.315	0.013	0.102	0.138	0.719
	Lasso	0.013	0.102	0.138	0.314	0.012	0.181	0.226	0.876
	Ridge	0.013	0.054	0.077	0.694	0.013	0.102	0.138	0.660
	Enet	0.013	0.102	0.138	0.314	0.012	0.181	0.226	0.876
	Bagging	0.014	0.008	0.013	0.786	0.013	0.075	0.104	0.723
	RF	0.015	0.005	0.009	0.732	0.013	0.038	0.056	0.647
	Boosting	0.015	0.003	0.014	0.194	0.014	0.026	0.040	0.29
	NN1	0.016	0	0.001	0.806	0.014	0.018	0.028	0.095
	NN2	0.015	0.002	0.004	0.384	0.014	0.018	0.028	0.051
	NN3	0.015	0.005	0.009	0.917	0.013	0.038	0.056	0.126
	NN4	0.015	0.005	0.009	0.917	0.013	0.054	0.077	0.041
	NN5	0.014	0.008	0.013	0.479	0.015	0.005	0.009	0.132

Table A5: Backtesting results of 1-day-ahead VaR for the CAC40 Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	Freq	RV5			RVTS			
			K	C	C-p	Freq	K	C	C-p
0.05	HAR	0.073	0	0	0.86	0.07	0	0	0.496
	HARX	0.069	0	0	0.525	0.067	0	0	0.502
	Lasso	0.068	0	0	0.627	0.066	0	0	0.62
	Ridge	0.067	0	0	0.378	0.066	0	0	0.248
	Enet	0.069	0	0	0.437	0.066	0	0	0.414
	Bagging	0.069	0	0	0.637	0.068	0	0	0.94
	RF	0.069	0	0	0.533	0.069	0	0	0.89
	Boosting	0.071	0	0	0.614	0.068	0	0	0.671
	NN1	0.069	0	0	0.613	0.067	0	0	0.638
	NN2	0.069	0	0	0.663	0.067	0	0	0.904
	NN3	0.069	0	0	0.677	0.067	0	0	0.985
	NN4	0.07	0	0	0.691	0.068	0	0	0.97
	NN5	0.069	0	0	0.988	0.068	0	0	0.932
0.01	HAR	0.03	0	0	0.774	0.032	0	0	0.983
	HARX	0.029	0	0	0.822	0.031	0	0	0.816
	Lasso	0.029	0	0	0.734	0.029	0	0	0.964
	Ridge	0.028	0	0	0.809	0.028	0	0	0.934
	Enet	0.029	0	0	0.734	0.029	0	0	0.964
	Bagging	0.028	0	0	0.99	0.03	0	0	0.844
	RF	0.028	0	0	0.495	0.03	0	0	0.766
	Boosting	0.03	0	0	0.842	0.031	0	0	0.781
	NN1	0.031	0	0	0.486	0.032	0	0	0.473
	NN2	0.029	0	0	0.88	0.031	0	0	0.527
	NN3	0.029	0	0	0.966	0.031	0	0	0.453
	NN4	0.029	0	0	0.966	0.031	0	0	0.672
	NN5	0.03	0	0	0.951	0.03	0	0	0.555

Table A6: Backtesting results of 1-day-ahead VaR for the DAX Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	RV5				RVTS			
		Freq	K	C	C-p	Freq	K	C	C-p
0.05	HAR	0.092	0	0	0.329	0.093	0	0	0.465
	HARX	0.092	0	0	0.086	0.091	0	0	0.087
	Lasso	0.09	0	0	0.039	0.09	0	0	0.036
	Ridge	0.085	0	0	0.048	0.086	0	0	0.037
	Enet	0.09	0	0	0.056	0.09	0	0	0.036
	Bagging	0.088	0	0	0.059	0.089	0	0	0.265
	RF	0.086	0	0	0.032	0.087	0	0	0.156
	Boosting	0.086	0	0	0.079	0.087	0	0	0.121
	NN1	0.09	0	0	0.074	0.091	0	0	0.077
	NN2	0.09	0	0	0.074	0.09	0	0	0.081
	NN3	0.09	0	0	0.107	0.089	0	0	0.158
	NN4	0.09	0	0	0.142	0.09	0	0	0.12
	NN5	0.09	0	0	0.075	0.091	0	0	0.062
0.01	HAR	0.046	0	0	0.053	0.045	0	0	0.095
	HARX	0.044	0	0	0.022	0.043	0	0	0.039
	Lasso	0.042	0	0	0.026	0.042	0	0	0.02
	Ridge	0.041	0	0	0.009	0.04	0	0	0.018
	Enet	0.042	0	0	0.02	0.041	0	0	0.037
	Bagging	0.042	0	0	0.158	0.04	0	0	0.389
	RF	0.042	0	0	0.106	0.039	0	0	0.158
	Boosting	0.043	0	0	0.045	0.041	0	0	0.076
	NN1	0.043	0	0	0.057	0.043	0	0	0.212
	NN2	0.045	0	0	0.081	0.043	0	0	0.167
	NN3	0.045	0	0	0.038	0.042	0	0	0.571
	NN4	0.045	0	0	0.052	0.042	0	0	0.548
	NN5	0.045	0	0	0.151	0.043	0	0	0.403

Table A7: Backtesting results of 1-day-ahead VaR for the EURO50 Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	Freq	RV5			RVTS			
			K	C	C-p	Freq	K	C	C-p
0.05	HAR	0.077	0	0	0.863	0.072	0	0	0.985
	HARX	0.07	0	0	0.392	0.068	0	0	0.381
	Lasso	0.07	0	0	0.439	0.067	0	0	0.314
	Ridge	0.068	0	0	0.524	0.065	0	0	0.607
	Enet	0.07	0	0	0.439	0.067	0	0	0.384
	Bagging	0.071	0	0	0.453	0.068	0	0	0.654
	RF	0.071	0	0	0.417	0.068	0	0	0.515
	Boosting	0.07	0	0	0.669	0.066	0	0	0.751
	NN1	0.069	0	0	0.282	0.066	0	0	0.294
	NN2	0.069	0	0	0.233	0.066	0	0	0.406
	NN3	0.068	0	0	0.236	0.066	0	0	0.505
	NN4	0.07	0	0	0.403	0.067	0	0	0.502
	NN5	0.069	0	0	0.534	0.067	0	0	0.745
0.01	HAR	0.033	0	0	0.904	0.034	0	0	0.943
	HARX	0.031	0	0	0.538	0.032	0	0	0.264
	Lasso	0.031	0	0	0.589	0.031	0	0	0.372
	Ridge	0.03	0	0	0.401	0.03	0	0	0.589
	Enet	0.031	0	0	0.531	0.031	0	0	0.345
	Bagging	0.031	0	0	0.245	0.03	0	0	0.159
	RF	0.029	0	0	0.357	0.03	0	0	0.12
	Boosting	0.031	0	0	0.148	0.031	0	0	0.218
	NN1	0.032	0	0	0.218	0.032	0	0	0.198
	NN2	0.032	0	0	0.357	0.032	0	0	0.576
	NN3	0.032	0	0	0.444	0.032	0	0	0.339
	NN4	0.031	0	0	0.481	0.032	0	0	0.612
	NN5	0.032	0	0	0.77	0.032	0	0	0.476

Table A8: Backtesting results of 1-day-ahead VaR for the FTSE100 Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	Freq	RV5			RVTS			
			K	C	C-p	Freq	K	C	C-p
0.05	HAR	0.071	0	0	0.617	0.081	0	0	0.471
	HARX	0.068	0	0	0.016	0.078	0	0	0.054
	Lasso	0.067	0	0	0.021	0.076	0	0	0.061
	Ridge	0.066	0	0	0.013	0.074	0	0	0.065
	Enet	0.067	0	0	0.016	0.075	0	0	0.056
	Bagging	0.068	0	0	0.033	0.075	0	0	0.125
	RF	0.067	0	0	0.035	0.075	0	0	0.106
	Boosting	0.071	0	0	0.055	0.079	0	0	0.28
	NN1	0.072	0	0	0.354	0.082	0	0	0.195
	NN2	0.071	0	0	0.082	0.081	0	0	0.111
	NN3	0.07	0	0	0.059	0.08	0	0	0.154
	NN4	0.069	0	0	0.023	0.08	0	0	0.079
	NN5	0.07	0	0	0.066	0.078	0	0	0.217
0.01	HAR	0.034	0	0	0.603	0.036	0	0	0.703
	HARX	0.03	0	0	0.08	0.033	0	0	0.341
	Lasso	0.029	0	0	0.133	0.032	0	0	0.221
	Ridge	0.029	0	0	0.093	0.032	0	0	0.037
	Enet	0.029	0	0	0.133	0.032	0	0	0.209
	Bagging	0.032	0	0	0.449	0.032	0	0	0.404
	RF	0.031	0	0	0.11	0.033	0	0	0.476
	Boosting	0.031	0	0	0.673	0.032	0	0	0.915
	NN1	0.033	0	0	0.736	0.034	0	0	0.842
	NN2	0.031	0	0	0.337	0.033	0	0	0.92
	NN3	0.031	0	0	0.479	0.033	0	0	0.778
	NN4	0.031	0	0	0.168	0.034	0	0	0.963
	NN5	0.032	0	0	0.498	0.033	0	0	0.779

Table A9: Backtesting results of 1-day-ahead VaR for the Nasdaq100 Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	Freq	RV5			RVTS			
			K	C	C-p	Freq	K	C	C-p
0.05	HAR	0.06	0.005	0.003	0.064	0.068	0	0	0.085
	HARX	0.062	0.001	0	0.133	0.068	0	0	0.036
	Lasso	0.061	0.002	0	0.235	0.068	0	0	0.034
	Ridge	0.057	0.034	0.007	0.518	0.063	0	0	0.038
	Enet	0.059	0.007	0.001	0.359	0.066	0	0	0.039
	Bagging	0.061	0.002	0	0.043	0.068	0	0	0.076
	RF	0.061	0.002	0	0.066	0.067	0	0	0.074
	Boosting	0.06	0.004	0	0.277	0.065	0	0	0.201
	NN1	0.064	0	0	0.213	0.07	0	0	0.064
	NN2	0.064	0	0	0.169	0.07	0	0	0.139
	NN3	0.062	0.001	0	0.183	0.069	0	0	0.096
	NN4	0.063	0	0	0.119	0.069	0	0	0.106
	NN5	0.063	0	0	0.171	0.069	0	0	0.253
0.01	HAR	0.029	0	0	0.299	0.031	0	0	0.994
	HARX	0.03	0	0	0.419	0.032	0	0	0.424
	Lasso	0.029	0	0	0.502	0.032	0	0	0.42
	Ridge	0.028	0	0	0.387	0.03	0	0	0.495
	Enet	0.029	0	0	0.559	0.031	0	0	0.476
	Bagging	0.03	0	0	0.439	0.031	0	0	0.136
	RF	0.028	0	0	0.258	0.031	0	0	0.272
	Boosting	0.027	0	0	0.443	0.029	0	0	0.933
	NN1	0.031	0	0	0.522	0.032	0	0	0.498
	NN2	0.032	0	0	0.909	0.033	0	0	0.67
	NN3	0.031	0	0	0.697	0.034	0	0	0.628
	NN4	0.032	0	0	0.436	0.034	0	0	0.74
	NN5	0.031	0	0	0.628	0.034	0	0	0.753

Table A10: Backtesting results of 1-day-ahead VaR for the NIFTY50 Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	Freq	RV5			RVTS			
			K	C	C-p	Freq	K	C	C-p
0.05	HAR	0.034	0	0	0.392	0.025	0	0	0.303
	HARX	0.031	0	0	0.403	0.024	0	0	0.187
	Lasso	0.029	0	0	0.145	0.022	0	0	0.198
	Ridge	0.028	0	0	0.212	0.021	0	0	0.086
	Enet	0.029	0	0	0.164	0.021	0	0	0.238
	Bagging	0.032	0	0	0.448	0.023	0	0	0.21
	RF	0.031	0	0	0.622	0.023	0	0	0.163
	Boosting	0.035	0	0	0.883	0.026	0	0	0.199
	NN1	0.034	0	0	0.495	0.026	0	0	0.151
	NN2	0.034	0	0	0.269	0.024	0	0	0.299
	NN3	0.032	0	0	0.316	0.023	0	0	0.247
	NN4	0.034	0	0	0.253	0.023	0	0	0.25
	NN5	0.034	0	0	0.148	0.023	0	0	0.243
0.01	HAR	0.003	0	0	0.002	0.001	0	0	0.867
	HARX	0.002	0	0	0.032	0.001	0	0	0.334
	Lasso	0.002	0	0	0.032	0.001	0	0	0.334
	Ridge	0.002	0	0	0.032	0.001	0	0	0.334
	Enet	0.002	0	0	0.032	0.001	0	0	0.334
	Bagging	0.002	0	0	0.023	0.001	0	0	0.867
	RF	0.002	0	0	0.023	0.001	0	0	0.867
	Boosting	0.002	0	0	0.101	0.001	0	0	0.82
	NN1	0.002	0	0	0.215	0	0	0	1
	NN2	0.002	0	0	0.029	0	0	0	1
	NN3	0.002	0	0	0.029	0	0	0	1
	NN4	0.002	0	0	0.26	0.001	0	0	0.334
	NN5	0.002	0	0	0.03	0.001	0	0	0.334

Table A11: Backtesting results of 1-day-ahead VaR for the NIKKEI225 Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	Freq	RV5			RVTS			
			K	C	C-p	Freq	K	C	C-p
0.05	HAR	0.086	0	0	0.404	0.082	0	0	0.573
	HARX	0.074	0	0	0.205	0.072	0	0	0.397
	Lasso	0.073	0	0	0.263	0.071	0	0	0.374
	Ridge	0.071	0	0	0.194	0.07	0	0	0.393
	Enet	0.073	0	0	0.255	0.072	0	0	0.358
	Bagging	0.076	0	0	0.441	0.076	0	0	0.588
	RF	0.077	0	0	0.4	0.074	0	0	0.64
	Boosting	0.076	0	0	0.329	0.075	0	0	0.795
	NN1	0.078	0	0	0.493	0.073	0	0	0.773
	NN2	0.075	0	0	0.503	0.073	0	0	0.772
	NN3	0.075	0	0	0.588	0.073	0	0	0.74
	NN4	0.073	0	0	0.886	0.072	0	0	0.747
	NN5	0.074	0	0	0.653	0.074	0	0	0.998
0.01	HAR	0.041	0	0	0.799	0.037	0	0	0.439
	HARX	0.032	0	0	0.587	0.032	0	0	0.803
	Lasso	0.03	0	0	0.596	0.031	0	0	0.645
	Ridge	0.03	0	0	0.66	0.03	0	0	0.587
	Enet	0.031	0	0	0.537	0.031	0	0	0.726
	Bagging	0.034	0	0	0.494	0.032	0	0	0.896
	RF	0.035	0	0	0.434	0.035	0	0	0.445
	Boosting	0.037	0	0	0.255	0.033	0	0	0.24
	NN1	0.034	0	0	0.374	0.032	0	0	0.298
	NN2	0.033	0	0	0.609	0.031	0	0	0.332
	NN3	0.032	0	0	0.196	0.032	0	0	0.361
	NN4	0.034	0	0	0.246	0.031	0	0	0.117
	NN5	0.034	0	0	0.276	0.032	0	0	0.208

Table A12: Backtesting results of 1-day-ahead VaR for the SCI Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	Freq	RV5			Freq	RVTS		
			K	C	C-p		K	C	C-p
0.05	HAR	0.054	0.285	0.206	0.055	0.047	0.445	0.549	0.877
	HARX	0.056	0.116	0.239	0.12	0.048	0.49	0.756	0.738
	Lasso	0.053	0.394	0.432	0.06	0.048	0.538	0.786	0.698
	Ridge	0.053	0.394	0.569	0.155	0.048	0.538	0.651	0.612
	Enet	0.054	0.319	0.477	0.085	0.048	0.538	0.786	0.724
	Bagging	0.052	0.479	0.663	0.069	0.047	0.36	0.576	0.506
	RF	0.055	0.153	0.35	0.027	0.049	0.695	0.769	0.382
	Boosting	0.058	0.027	0.088	0.033	0.05	0.924	0.986	0.727
	NN1	0.056	0.075	0.203	0.024	0.051	0.73	0.91	0.59
	NN2	0.056	0.116	0.184	0.003	0.05	0.982	0.973	0.478
	NN3	0.056	0.116	0.278	0.026	0.049	0.865	0.971	0.559
	NN4	0.056	0.087	0.181	0.024	0.049	0.695	0.924	0.704
	NN5	0.056	0.116	0.278	0.039	0.049	0.808	0.962	0.579
0.01	HAR	0.027	0	0	0.848	0.023	0	0	0.019
	HARX	0.027	0	0	0.643	0.024	0	0	1
	Lasso	0.027	0	0	0.866	0.023	0	0	0.147
	Ridge	0.026	0	0	0.889	0.023	0	0	0.165
	Enet	0.027	0	0	0.97	0.023	0	0	0.147
	Bagging	0.026	0	0	0.33	0.024	0	0	0.108
	RF	0.026	0	0	0.644	0.024	0	0	0.109
	Boosting	0.03	0	0	0.395	0.026	0	0	0.118
	NN1	0.029	0	0	0.846	0.027	0	0	0.011
	NN2	0.028	0	0	0.911	0.026	0	0	0.019
	NN3	0.028	0	0	0.65	0.025	0	0	0.04
	NN4	0.029	0	0	0.82	0.025	0	0	0.07
	NN5	0.029	0	0	0.808	0.025	0	0	0.018

Table A13: Backtesting results of 1-day-ahead VaR for the S&P500 Index. K , C and CP stands for Kupiec, Christoffersen and Christoffersen-Pelletier p-values, respectively

α	Method	RV5				RVTS			
		<i>freq</i>	K	C	CP	<i>freq</i>	K	C	CP
0.05	HAR	0.074	0	0	0.369	0.075	0	0	0.223
	HARX	0.074	0	0	0.065	0.075	0	0	0.073
	Lasso	0.073	0	0	0.33	0.072	0	0	0.124
	Ridge	0.071	0	0	0.389	0.070	0	0	0.383
	Enet	0.072	0	0	0.265	0.072	0	0	0.121
	Bagging	0.072	0	0	0.576	0.072	0	0	0.572
	RF	0.071	0	0	0.741	0.070	0	0	0.909
	Boosting	0.075	0	0	0.364	0.073	0	0	0.952
	NN1	0.075	0	0	0.356	0.076	0	0	0.272
	NN2	0.075	0	0	0.475	0.075	0	0	0.368
	NN3	0.075	0	0	0.618	0.075	0	0	0.173
	NN4	0.074	0	0	0.544	0.075	0	0	0.278
	NN5	0.074	0	0	0.897	0.074	0	0	0.605
0.01	HAR	0.040	0	0	0.420	0.036	0	0	0.866
	HARX	0.042	0	0	0.721	0.038	0	0	0.602
	Lasso	0.040	0	0	0.777	0.036	0	0	0.557
	Ridge	0.039	0	0	0.857	0.036	0	0	0.594
	Enet	0.039	0	0	0.730	0.035	0	0	0.481
	Bagging	0.040	0	0	0.877	0.036	0	0	0.523
	RF	0.039	0	0	0.992	0.036	0	0	0.520
	Boosting	0.041	0	0	0.852	0.038	0	0	0.899
	NN1	0.041	0	0	0.452	0.038	0	0	0.808
	NN2	0.041	0	0	0.451	0.038	0	0	0.766
	NN3	0.040	0	0	0.478	0.038	0	0	0.914
	NN4	0.040	0	0	0.798	0.038	0	0	0.981
	NN5	0.040	0	0	0.638	0.037	0	0	0.931